

Lab 1: Exploring Insecure Cameras and OSINT Techniques

Introduction to Cybersecurity

Hands-On Security Research Lab

Lab Information

Duration: 2-3 hours

Difficulty Level: Beginner

Lab Type: Guided reconnaissance exercise

Tools Required: Web browser, Shodan account (free)

Table of Contents

Contents

Table of Contents	2
1. Lab Overview.....	4
Introduction	4
What You'll Learn	4
Real-World Context	4
Why This Matters	4
2. Learning Objectives	5
Technical Skills	5
Professional Skills	5
Conceptual Understanding	5
3. Prerequisites and Setup	6
Knowledge Requirements	6
Technical Requirements	6
Pre-Lab Setup Instructions	6
Important Safety Reminders.....	7
4. Technical Background	8
What Are Insecure Cameras?	8
Why Are These Devices Exposed?	8
What is OSINT?	8
Legal and Ethical Boundaries.....	9
5. Lab Environment	10
Required Tools:	10
6. Exercise 1: Google Dorks for Camera Discovery.....	11
Objective	11
What Are Google Dorks?	11
Exercise Steps	11
Search 1: Webcam XP Software.....	11
Search 2: Network Camera Directory Listings.....	11
Search 3: Live View Webcams (Panasonic).....	12
Search 4: Axis Network Cameras	12
Search 5: MJPEG Stream Cameras	12
Additional Searches to Try:	12
Documentation Template	12
Exercise 1 Reflection Questions.....	12
7. Exercise 2: Shodan Reconnaissance	13

Objective	13
What is Shodan?.....	13
Getting Started with Shodan.....	13
Basic Shodan Searches	13
Search 1: Generic Webcams.....	13
Search 2: Default Passwords	13
Search 3: Cameras by Port.....	14
Search 4: Hikvision Cameras.....	14
Search 5: RTSP Streaming Protocol	14
Advanced Shodan Filters	14
Using Shodan Maps	14
8. Exercise 3: Default Credentials Research	16
9. Exercise 4: Alternative OSINT Tools	17
10. Exercise 5: Documenting and Reporting Findings	18
11. Lab Reflection Questions.....	19
12. Additional Challenges (Optional)	20
13. Cleanup and Submission.....	21
Cleanup Steps:.....	21
Submission Requirements:	21
14. Additional Resources	22
15. Instructor Notes.....	23
Time Management:	23
Common Student Issues:	23
Assessment Rubric:	23
Lab Completion Checklist.....	23

1. Lab Overview

Introduction

Welcome to your first hands-on cybersecurity lab! In this exercise, you'll learn the fundamental techniques that security professionals use to discover and analyze internet-connected devices. You'll explore how millions of cameras, webcams, and other IoT devices are unintentionally exposed to the public internet, creating serious security and privacy risks.

This lab focuses on passive reconnaissance—observing and documenting what's publicly accessible without attempting to access, modify, or interact with any discovered systems. These are the exact same techniques used by security researchers, penetration testers, and unfortunately, malicious actors.

What You'll Learn

Throughout this lab, you will:

- Master advanced Google search operators (Google dorks) to discover exposed devices
- Use Shodan, the world's most powerful device search engine, to map internet-connected cameras
- Research common default credentials that make devices vulnerable to unauthorized access
- Explore alternative OSINT (Open Source Intelligence) platforms and tools
- Document your findings professionally, following industry-standard reporting practices
- Understand the legal and ethical boundaries of security research

Real-World Context

The techniques you'll practice in this lab are used daily by:

- Security researchers discovering vulnerabilities before malicious actors do
- Penetration testers conducting authorized security assessments
- Threat intelligence analysts tracking attacker infrastructure
- Bug bounty hunters finding and responsibly disclosing security flaws
- Security operations teams identifying shadow IT and unauthorized devices

Why This Matters

According to recent security research:

- Over 15 million webcams and IP cameras are exposed on the public internet
- 73% of IoT devices use default or weak credentials
- The average exposed camera remains vulnerable for 8+ months before being secured
- 60% of organizations don't know how many internet-connected devices they have
- Insecure cameras have been used in massive botnet attacks (like the Mirai botnet in 2016)

2. Learning Objectives

By the end of this lab, you will be able to:

Technical Skills

1. **Construct effective Google dork queries** using advanced search operators (intitle:, inurl:, filetype:) to discover exposed devices and sensitive information
2. **Navigate and utilize Shodan** to search for internet-connected cameras, understanding filters, search syntax, and result interpretation
3. **Research default credentials** for common IoT devices and understand why they represent critical security vulnerabilities
4. **Explore alternative OSINT platforms** such as Censys, ZoomEye, and Wigle to complement your reconnaissance capabilities
5. **Analyze and interpret device banners**, service information, and metadata to assess security posture

Professional Skills

6. **Document reconnaissance findings** in a clear, professional format suitable for security reports and stakeholder communication
7. **Practice responsible disclosure** procedures for discovered vulnerabilities, following industry-standard ethical guidelines
8. **Create professional security assessments** with clear findings, evidence, and actionable recommendations

Conceptual Understanding

9. **Understand how devices become exposed** on the internet through misconfiguration, lack of security awareness, and default settings
10. **Recognize the security implications** of exposed IoT devices for individuals, organizations, and critical infrastructure
11. **Distinguish between legal and illegal activities** in security research, understanding laws like the Computer Fraud and Abuse Act (CFAA)
12. **Apply ethical frameworks** to security research decisions, balancing curiosity with responsibility

3. Prerequisites and Setup

Knowledge Requirements

This lab is designed for beginners, but you should have:

- • Basic understanding of IP addresses and how the internet works
- • Familiarity with using web browsers and search engines
- • Basic computer skills (file management, note-taking)
- • Interest in cybersecurity and willingness to learn

Technical Requirements

Hardware:

- • Computer with internet connection (desktop or laptop)
- • Minimum 4GB RAM recommended
- • Screen resolution 1366x768 or higher for optimal viewing

Software:

- • Modern web browser (Chrome, Firefox, or Edge - latest version)
- • Note-taking application (Microsoft Word, Google Docs, or Notepad)
- • PDF reader for saving web pages and documentation
- • Screenshot tool (built-in: Snipping Tool for Windows, Cmd+Shift+4 for Mac)

Accounts:

- • Shodan account (free tier) - Create at <https://www.shodan.io>
- • Email address for account verification

Pre-Lab Setup Instructions

Complete these steps BEFORE starting the lab exercises:

Step 1: Create Shodan Account

- a) Navigate to <https://www.shodan.io>
- b) Click 'Sign Up' in the top right corner
- c) Enter your email address and create a strong password
- d) Verify your email address via the confirmation link
- e) Log in to confirm your account is active

Step 2: Prepare Documentation Template

- a) Create a new Word document or Google Doc
- b) Title it: 'Lab1_OSINT_YourLastName'
- c) Include your name, date, and course information at the top
- d) Create sections for each exercise

Step 3: Test Your Screenshot Tool

- a) Take a practice screenshot of your desktop
- b) Ensure you can save and insert it into your document
- c) Practice annotating screenshots if needed (arrows, text boxes)

Important Safety Reminders

⚠ CRITICAL: This lab involves PASSIVE reconnaissance ONLY

You must NOT:

- **✗** Attempt to login to any discovered systems
- **✗** Test default credentials on live systems
- **✗** Exploit any vulnerabilities you discover
- **✗** Access, download, or modify any data
- **✗** Share or distribute camera feeds or device information publicly
- **✗** Attempt to contact device owners without authorization

Viewing publicly accessible information is legal. Accessing systems without authorization is a federal crime under the Computer Fraud and Abuse Act (CFAA) and can result in criminal prosecution, fines, and imprisonment.

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4. Technical Background

What Are Insecure Cameras?

IP cameras and webcams are network-connected devices that stream video over the internet. While incredibly useful for security monitoring, remote work, and home surveillance, many of these devices suffer from critical security vulnerabilities that make them accessible to anyone on the internet.

Common Security Issues:

- **Default credentials never changed:** Manufacturers ship devices with usernames like 'admin' and passwords like 'admin' or '12345'. Many users never change these defaults.
- **No authentication required:** Some cameras allow anyone to view streams without any login at all.
- **Outdated firmware:** Cameras with known vulnerabilities that were never patched.
- **Exposed administrative interfaces:** Configuration pages accessible from the internet without VPN protection.
- **Misconfigured port forwarding:** Users enable remote access without proper security measures.
- **Lack of encryption:** Video streams transmitted in clear text without HTTPS/TLS.

Why Are These Devices Exposed?

Devices become discoverable through several mechanisms:

1. **Search Engine Indexing:** Google and other search engines automatically crawl and index web-accessible devices. If a camera's web interface is accessible via HTTP/HTTPS without authentication, search engines will index it.
2. **Port Scanning:** Tools like Shodan actively scan the entire IPv4 address space, probing common ports (80, 443, 554, 8080, 8081) and cataloging what services respond.
3. **User Misconfiguration:** Users enable remote access features (UPnP, port forwarding) to view cameras while away from home, often without understanding the security implications.
4. **Default Settings:** Manufacturers prioritize ease-of-use over security, shipping devices in insecure default configurations. The expectation that users will secure them is often unrealistic.

What is OSINT?

Open Source Intelligence (OSINT) refers to collecting and analyzing information from publicly available sources. Unlike classified intelligence gathering, OSINT uses only legal, public sources such as:

- Search engines (Google, Bing)
- Social media platforms (LinkedIn, Twitter, Facebook)
- Public records and databases
- Company websites and press releases
- Technical forums and code repositories

- • Device search engines (Shodan, Censys)

In cybersecurity, OSINT techniques are used for:

- • Reconnaissance during penetration testing
- • Threat intelligence gathering (tracking threat actors)
- • Security research and vulnerability discovery
- • Incident response investigations
- • Competitive intelligence

Legal and Ethical Boundaries

Understanding the legal and ethical implications of security research is critical:

LEGAL Activities:

- ☒ Viewing publicly accessible information (no authentication required)
- ☒ Using search engines and OSINT tools
- ☒ Reading publicly posted data
- ☒ Documenting findings for educational purposes
- ☒ Responsible disclosure to affected parties

ILLEGAL Activities:

- ☒ Accessing systems without authorization (even with default credentials)
- ☒ Using discovered credentials to login
- ☒ Exploiting vulnerabilities on systems you don't own
- ☒ Interfering with systems or services (DoS attacks)
- ☒ Downloading or exfiltrating data
- ☒ Sharing private camera feeds publicly

The Computer Fraud and Abuse Act (CFAA) makes it a federal crime to access a computer without authorization or to exceed authorized access. Penalties can include fines up to \$250,000 and imprisonment up to 20 years for repeat offenses.

5. Lab Environment

This lab uses entirely web-based tools that require no special software installation. All exercises can be completed using your web browser and the accounts you created during setup.

Required Tools:

1. Web Browser

- Purpose: Primary interface for all lab activities
- Configuration: No special settings required
- Recommendation: Use private/incognito mode to avoid rate limiting

2. Shodan (<https://www.shodan.io>)

- Purpose: Internet-connected device search engine
- Account: Free tier (provides 100 search results per month)
- Features: Device discovery, banner analysis, geographical mapping

3. Google Search

- Purpose: Advanced search operators for device discovery
- Account: Not required
- Note: May rate-limit excessive searches; use responsibly

4. Documentation Tool

- Purpose: Recording findings and creating lab report
- Options: Microsoft Word, Google Docs, Notion, or simple text editor

6. Exercise 1: Google Dorks for Camera Discovery

Objective

Learn to use advanced Google search operators (dorks) to discover exposed cameras and devices. Google dorks are powerful search queries that use special operators to find specific types of content that normal searches might miss.

What Are Google Dorks?

Google dorks are advanced search queries that use special operators to find specific information. These operators tell Google to search in specific locations or for specific types of content.

Common Google Search Operators:

intitle: - Search for keywords in page titles

inurl: - Search for keywords in URLs

filetype: - Search for specific file types (PDF, DOC, XLS)

site: - Limit search to specific domain or website

"exact phrase" - Search for exact text matches (must be in quotes)

Exercise Steps

Search 1: Webcam XP Software

Google Dork: `intitle:"index of" "xp cam 5"`

What this finds: Webcam XP is popular webcam software that often exposes camera feeds through directory listings. This search looks for directory indexes ('index of') containing the text 'xp cam 5'.

Instructions:

- 1. Open your web browser and navigate to google.com
- 2. Enter the search query exactly as shown: `intitle:"index of" "xp cam 5"`
- 3. Review the results - look for directory listings showing camera feeds
- 4. Take a screenshot of your search results
- 5. Document your findings (number of results, countries, camera types)

[Screenshot Placeholder: Insert screenshot of Google search results here]

Analysis Questions:

- • How many results did you find?
- • What countries are these cameras located in?
- • Can you identify what types of locations these cameras monitor?
- • Do any results require authentication, or are they completely open?

Search 2: Network Camera Directory Listings

Google Dork: `intitle:"index of" inurl:ftp "network camera"`

What this finds: FTP directories containing network camera files and configurations. This combines three operators: intitle for 'index of', inurl for 'ftp' in the URL, and exact phrase match for 'network camera'.

Execute the search, take screenshots, and document your findings using the same process as Search 1.

Search 3: Live View Webcams (Panasonic)

Google Dork: `inurl:view/view.shtml`

What this finds: Panasonic network cameras with default accessible live view pages. The URL pattern 'view/view.shtml' is characteristic of Panasonic IP cameras.

Search 4: Axis Network Cameras

Google Dork: `inurl:/view/index.shtml`

What this finds: Axis brand IP cameras with default web interfaces. Axis is one of the largest manufacturers of professional network cameras.

Search 5: MJPEG Stream Cameras

Google Dork: `inurl:mjpg/video.mjpg`

What this finds: Motion JPEG video streams from various IP cameras. MJPEG is a common streaming format that displays a series of JPEG images.

Additional Searches to Try:

Search 6: `intitle:"DVR Login" OR intitle:"Network Video Recorder"`

Search 7: `intitle:"Webcam 7" OR intitle:"webcamXP 5"`

Search 8: `inurl:indexFrame.shtml OR inurl:cgiUserName`

Documentation Template

For each search, document the following:

Search Query	Results Found	Camera Type	Authentication Required?	Notes
<code>intitle:"index of" "xp cam 5"</code>				
<code>inurl:view/view.shtml</code>				

Exercise 1 Reflection Questions

Answer these questions in your lab report:

- 1. Which Google dork returned the most results? Why do you think this is?
- 2. What percentage of discovered cameras required authentication vs. being completely open?
- 3. What patterns did you notice in the URLs or page titles of exposed cameras?
- 4. Why do you think these devices are indexed by Google in the first place?
- 5. If you were advising a friend who just installed an IP camera, what security recommendations would you give them based on what you've discovered?

7. Exercise 2: Shodan Reconnaissance

Objective

Use Shodan to discover and analyze internet-connected cameras and devices. Learn to interpret device information, use search filters, and understand the information Shodan provides about each discovered device.

What is Shodan?

Shodan is a search engine that continuously scans the entire internet for connected devices. Unlike Google (which indexes web content), Shodan indexes:

- Open ports on IP addresses
- Running services and their versions
- Device banners (service identification information)
- Known vulnerabilities
- Geographical locations
- Screenshots of HTTP services

Shodan is often called "the world's most dangerous search engine" because it reveals devices that should not be publicly accessible, including industrial control systems, building management systems, power plants, and yes—millions of webcams and IP cameras.

Getting Started with Shodan

Step 1: Log into your Shodan account at <https://www.shodan.io>

Step 2: Review your account dashboard to see your search quota (free accounts get 100 results per month)

Step 3: Familiarize yourself with the search interface

Basic Shodan Searches

Search 1: Generic Webcams

Shodan Query: `webcam`

This simple search finds devices identified as webcams. Review the results to understand what information Shodan provides:

- IP address and hostname
- Geographical location (country, city, coordinates)
- ISP (Internet Service Provider)
- Open ports and services
- Device banner information
- Screenshots (if available)

Search 2: Default Passwords

Shodan Query: `"default password"`

This search finds devices that advertise their default passwords in their banner information—a serious security vulnerability.

Search 3: Cameras by Port

Shodan Query: `port:8081 camera`

Cameras commonly run on specific ports. Port 8081 is frequently used for camera web interfaces.

Search 4: Hikvision Cameras

Shodan Query: `"Hikvision-Webs"`

Hikvision is one of the world's largest surveillance camera manufacturers. This search finds Hikvision branded cameras by their distinctive web server banner.

Search 5: RTSP Streaming Protocol

Shodan Query: `port:554 has_screenshot:true`

RTSP (Real-Time Streaming Protocol) on port 554 is commonly used for IP camera streaming. The `has_screenshot` filter shows only results where Shodan captured a preview image.

Advanced Shodan Filters

Shodan supports powerful filters for precise searching. Here are the most useful ones:

Filter	Purpose	Example
<code>port:</code>	Search specific port	<code>port:80</code>
<code>country:</code>	Filter by country code	<code>country:"CA"</code>
<code>city:</code>	Filter by city	<code>city:"Toronto"</code>
<code>has_screenshot:</code>	Only results with screenshots	<code>has_screenshot:true</code>
<code>vuln:</code>	Devices with specific CVE	<code>vuln:CVE-2017-12611</code>

Try combining multiple filters:

Example: `webcam country:"CA" has_screenshot:true port:8080`

This searches for webcams in Canada with screenshots available running on port 8080.

Using Shodan Maps

Shodan includes a geographical mapping feature:

- 1. Perform any search
- 2. Click the 'Maps' tab in the results
- 3. View the geographical distribution of discovered devices
- 4. Zoom in to see device clustering
- 5. Click individual markers for device details

Analysis Questions:

- • Which regions have the highest concentration of exposed cameras?

- • Why do you think certain areas have more exposed devices?
- • What does this tell ou bout global cybersecurity awareness?

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8. Exercise 3: Default Credentials Research

Research common default credentials for popular IP camera brands using public databases. Document your findings and understand why default credentials represent critical vulnerabilities. DO NOT attempt to use these credentials on any live systems.

Resources:

- • DefaultCreds-cheat-sheet (GitHub)
- • CIRT.net Default Password Database
- • RouterPasswords.com

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9. Exercise 4: Alternative OSINT Tools

Explore additional reconnaissance tools:

- Censys (<https://search.censys.io>) - Similar to Shodan with certificate focus
- ZoomEye (<https://www.zoomeye.org>) - Chinese alternative
- Wigle (<https://wigle.net>) - Wireless network mapping

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10. Exercise 5: Documenting and Reporting Findings

Create a professional OSINT reconnaissance report including:

- Executive summary
- Methodology (tools used, searches performed)
- Findings (categorized by severity)
- Statistics and analysis
- Recommendations
- Evidence (screenshots)

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11. Lab Reflection Questions

Answer the following in your lab report:

Technical Questions:

- 1. What percentage of discovered cameras required authentication?
- 2. Which Google dork was most effective?
- 3. What are the top 3 ports commonly used by IP cameras?
- 4. Name three manufacturers whose devices you discovered.
- 5. What is the difference between Shodan and Google for device discovery?

Analysis Questions:

- 6. Why are so many cameras exposed on the internet?
- 7. What steps should device owners take to secure cameras?
- 8. How could an attacker use the information gathered in this lab?
- 9. What role do manufacturers play in IoT security?
- 10. Should there be regulations for IoT device security? Why or why not?

Ethical Questions:

- 11. Where is the line between security research and unauthorized access?
- 12. If you discovered a camera in a sensitive location, what should you do?
- 13. Is viewing publicly accessible camera feeds unethical even without authentication bypass?
- 14. How would you feel if your camera was discovered in a search?

12. Additional Challenges (Optional)

For advanced students who want to deepen their skills:

- Challenge 1: Find and document 5 different types of IoT devices (not cameras)
- Challenge 2: Research vulnerabilities for one manufacturer and document CVEs
- Challenge 3: Create a security guide for consumers on securing IP cameras
- Challenge 4: Research and test command-line OSINT tools (theHarvester, Recon-ng)

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13. Cleanup and Submission

Cleanup Steps:

- 1. Clear browser history (optional, for privacy)
- 2. Log out of Shodan and other OSINT platforms
- 3. Close all browser tabs with discovered devices
- 4. DO NOT save or bookmark links to exposed cameras
- 5. Review documentation to ensure no sensitive info is included

Submission Requirements:

Your lab report must include:

- 1. Cover page (name, date, lab title)
- 2. Completed exercises with screenshots
- 3. Findings report from Exercise 5
- 4. All reflection questions answered
- 5. Documentation table of discoveries (anonymized)

File format: PDF

Naming: LastName_FirstName_Lab1.pdf

14. Additional Resources

Books:

- • 'Open Source Intelligence Techniques' by Michael Bazzell
- • 'The Hacker Playbook 3' by Peter Kim
- • 'OSINT Tools and Techniques' by Nihad Hassan

Online Courses:

- • SANS SEC487: Open-Source Intelligence Gathering
- • Udemy: OSINT Fundamentals
- • Cybrary: OSINT Training

Tools:

- • OSINT Framework (osintframework.com)
- • Awesome OSINT (GitHub)
- • Maltego (Community Edition)

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15. Instructor Notes

Time Management:

- Exercise 1 (Google Dorks): 30-40 minutes
- Exercise 2 (Shodan): 45-60 minutes
- Exercise 3 (Default Credentials): 15-20 minutes
- Exercise 4 (Alternative Tools): 20-25 minutes
- Exercise 5 (Documentation): 30-40 minutes
- Reflection and cleanup: 15-20 minutes

Common Student Issues:

- Shodan free account limits - remind students to use searches wisely
- Google rate limiting - suggest incognito mode or VPN if needed
- Ethical concerns - address early and often
- Over-curiosity - emphasize passive observation only

Assessment Rubric:

- Excellent (90-100%): All exercises complete, detailed screenshots, comprehensive report, thoughtful reflection
- Good (80-89%): Most exercises complete, adequate documentation, basic analysis
- Satisfactory (70-79%): Core exercises done, minimal documentation
- Needs Improvement (<70%): Incomplete exercises, missing documentation

Lab Completion Checklist

- ☐ Created Shodan account
- ☐ Completed all Google dork searches
- ☐ Performed Shodan reconnaissance
- ☐ Researched default credentials
- ☐ Explored alternative OSINT tools
- ☐ Created professional findings report
- ☐ Answered all reflection questions
- ☐ Included required screenshots
- ☐ Reviewed ethical guidelines
- ☐ Formatted report properly
- ☐ Submitted PDF document

End of Lab 1

Next Lab: Network Scanning and Enumeration